

Category Credit Detail			
Sl.No.	Description	Credits	Maximum Credit
1	FC - Foundation Core	53	53
2	DLES - Discipline-linked Engineering Sciences	12	12
3	DC - Discipline Core	47	47
4	DE - Discipline Elective	15	15
5	PI - Projects and Internship	9	9
6	OE - Open Elective	15	15
7	BC - Bridge Course	0	0
8	NGCR - Non-graded Core Requirement	11	11
9	ME - Multidisciplinary Elective	0	0
Total Credits		162	

Foundation Core									
sl.no	Course Code	Course Title	Course Type	Ver sio n	L	T	P	J	Credits
1	BCHY101L	Engineering Chemistry	Theory Only	1.0	3	0	0	0	3.0
2	BCHY101P	Engineering Chemistry Lab	Lab Only	1.0	0	0	2	0	1.0
3	BCSE101E	Computer Programming: Python	Embedded Theory and Lab	1.0	1	0	4	0	3.0
4	BCSE102L	Structured and Object-Oriented Programming	Theory Only	1.0	2	0	0	0	2.0
5	BCSE102P	Structured and Object-Oriented Programming Lab	Lab Only	1.0	0	0	4	0	2.0
6	BCSE103E	Computer Programming: Java	Embedded Theory and Lab	1.0	1	0	4	0	3.0
7	BEEE102L	Basic Electrical and Electronics Engineering	Theory Only	1.0	3	0	0	0	3.0
8	BEEE102P	Basic Electrical and Electronics Engineering Lab	Lab Only	1.0	0	0	2	0	1.0
9	BENG101L	Technical English Communication	Theory Only	1.0	2	0	0	0	2.0
10	BENG101P	Technical English Communication Lab	Lab Only	1.0	0	0	2	0	1.0
11	BENG102P	Technical Report Writing	Lab Only	1.0	0	0	2	0	1.0
12	BFLE200L	B.Tech. Foreign Language - 2021onwards	Basket	1.0	0	0	0	0	2.0
13	BHSM200L	B.Tech. HSM Elective - 2021 onwards	Basket	1.0	0	0	0	0	3.0
14	BMAT101L	Calculus	Theory Only	1.0	3	0	0	0	3.0
15	BMAT101P	Calculus Lab	Lab Only	1.0	0	0	2	0	1.0
16	BMAT102L	Differential Equations and Transforms	Theory Only	1.0	3	1	0	0	4.0
17	BMAT201L	Complex Variables and Linear Algebra	Theory Only	1.0	3	1	0	0	4.0
18	BMAT202L	Probability and Statistics	Theory Only	1.0	3	0	0	0	3.0
19	BMAT202P	Probability and Statistics Lab	Lab Only	1.0	0	0	2	0	1.0
20	BPHY101L	Engineering Physics	Theory Only	1.0	3	0	0	0	3.0
21	BPHY101P	Engineering Physics Lab	Lab Only	1.0	0	0	2	0	1.0
22	BSTS101P	Quantitative Skills Practice I	Soft Skill	1.0	0	0	3	0	1.5
23	BSTS102P	Quantitative Skills Practice II	Soft Skill	1.0	0	0	3	0	1.5
24	BSTS201P	Qualitative Skills Practice I	Soft Skill	1.0	0	0	3	0	1.5
25	BSTS202P	Qualitative Skills Practice II	Soft Skill	1.0	0	0	3	0	1.5

Discipline-linked Engineering Sciences									
sl.no	Course Code	Course Title	Course Type	Version	L	T	P	J	Credits
1	BECE102L	Digital Systems Design	Theory Only	1.0	3	0	0	0	3.0
2	BECE102P	Digital Systems Design Lab	Lab Only	1.0	0	0	2	0	1.0
3	BECE204L	Microprocessors and Microcontrollers	Theory Only	1.0	3	0	0	0	3.0
4	BECE204P	Microprocessors and Microcontrollers Lab	Lab Only	1.0	0	0	2	0	1.0
5	BMAT205L	Discrete Mathematics and Graph Theory	Theory Only	1.0	3	1	0	0	4.0

Discipline Core									
sl.no	Course Code	Course Title	Course Type	Version	L	T	P	J	Credits
1	BCSE202L	Data Structures and Algorithms	Theory Only	1.0	3	0	0	0	3.0
2	BCSE202P	Data Structures and Algorithms Lab	Lab Only	1.0	0	0	2	0	1.0
3	BCSE203E	Web Programming	Embedded Theory and Lab	1.0	1	0	4	0	3.0
4	BCSE204L	Design and Analysis of Algorithms	Theory Only	1.0	3	0	0	0	3.0
5	BCSE204P	Design and Analysis of Algorithms Lab	Lab Only	1.0	0	0	2	0	1.0
6	BCSE205L	Computer Architecture and Organization	Theory Only	1.0	3	0	0	0	3.0
7	BCSE301L	Software Engineering	Theory Only	1.0	3	0	0	0	3.0
8	BCSE301P	Software Engineering Lab	Lab Only	1.0	0	0	2	0	1.0
9	BCSE302L	Database Systems	Theory Only	1.0	3	0	0	0	3.0
10	BCSE302P	Database Systems Lab	Lab Only	1.0	0	0	2	0	1.0
11	BCSE303L	Operating Systems	Theory Only	1.0	3	0	0	0	3.0
12	BCSE303P	Operating Systems Lab	Lab Only	1.0	0	0	2	0	1.0
13	BCSE304L	Theory of Computation	Theory Only	1.0	3	0	0	0	3.0
14	BCSE305L	Embedded Systems	Theory Only	1.0	3	0	0	0	3.0
15	BCSE306L	Artificial Intelligence	Theory Only	1.0	3	0	0	0	3.0
16	BCSE307L	Compiler Design	Theory Only	1.0	3	0	0	0	3.0
17	BCSE307P	Compiler Design Lab	Lab Only	1.0	0	0	2	0	1.0
18	BCSE308L	Computer Networks	Theory Only	1.0	3	0	0	0	3.0
19	BCSE308P	Computer Networks Lab	Lab Only	1.0	0	0	2	0	1.0
20	BCSE309L	Cryptography and Network Security	Theory Only	1.0	3	0	0	0	3.0
21	BCSE309P	Cryptography and Network Security Lab	Lab Only	1.0	0	0	2	0	1.0

Discipline Elective									
sl.no	Course Code	Course Title	Course Type	Version	L	T	P	J	Credits
1	BCSE206L	Foundations of Data Science	Theory Only	1.0	3	0	0	0	3.0
2	BCSE207L	Programming for Data Science	Theory Only	1.0	2	0	0	0	2.0
3	BCSE207P	Programming for Data Science Lab	Lab Only	1.0	0	0	2	0	1.0

Discipline Elective									
4	BCSE208L	Data Mining	Theory Only	1.0	2	0	0	0	2.0
5	BCSE208P	Data Mining Lab	Lab Only	1.0	0	0	2	0	1.0
6	BCSE209L	Machine Learning	Theory Only	1.0	3	0	0	0	3.0
7	BCSE209P	Machine Learning Lab	Lab Only	1.0	0	0	2	0	1.0
8	BCSE310L	IoT Architectures and Protocols	Theory Only	1.0	3	0	0	0	3.0
9	BCSE311L	Sensors and Actuator Devices	Theory Only	1.0	2	0	0	0	2.0
10	BCSE311P	Sensors and Actuator Devices Lab	Lab Only	1.0	0	0	2	0	1.0
11	BCSE312L	Programming for IoT Boards	Theory Only	1.0	2	0	0	0	2.0
12	BCSE312P	Programming for IoT Boards Lab	Lab Only	1.0	0	0	2	0	1.0
13	BCSE313L	Fundamentals of Fog and Edge Computing	Theory Only	1.0	3	0	0	0	3.0
14	BCSE314L	Privacy and Security in IoT	Theory Only	1.0	3	0	0	0	3.0
15	BCSE315L	Wearable Computing	Theory Only	1.0	3	0	0	0	3.0
16	BCSE316L	Design of Smart Cities	Theory Only	1.0	3	0	0	0	3.0
17	BCSE317L	Information Security	Theory Only	1.0	3	0	0	0	3.0
18	BCSE318L	Data Privacy	Theory Only	1.0	3	0	0	0	3.0
19	BCSE319L	Penetration Testing and Vulnerability Analysis	Theory Only	1.0	2	0	0	0	2.0
20	BCSE319P	Penetration Testing and Vulnerability Analysis Lab	Lab Only	1.0	0	0	2	0	1.0
21	BCSE320L	Web Application Security	Theory Only	1.0	3	0	0	0	3.0
22	BCSE321L	Malware Analysis	Theory Only	1.0	2	0	0	0	2.0
23	BCSE321P	Malware Analysis Lab	Lab Only	1.0	0	0	2	0	1.0
24	BCSE322L	Digital Forensics	Theory Only	1.0	2	0	0	0	2.0
25	BCSE322P	Digital Forensics Lab	Lab Only	1.0	0	0	2	0	1.0
26	BCSE323L	Digital Watermarking and Steganography	Theory Only	1.0	3	0	0	0	3.0
27	BCSE324L	Foundations of Blockchain Technology	Theory Only	1.0	3	0	0	0	3.0
28	BCSE325L	Introduction to Bitcoin	Theory Only	1.0	3	0	0	0	3.0
29	BCSE326L	Blockchain Architecture Design	Theory Only	1.0	3	0	0	0	3.0
30	BCSE327L	Smart Contracts	Theory Only	1.0	2	0	0	0	2.0
31	BCSE327P	Smart Contracts Lab	Lab Only	1.0	0	0	2	0	1.0
32	BCSE328L	Cryptocurrency Technologies	Theory Only	1.0	3	0	0	0	3.0
33	BCSE329L	Blockchain and Distributed Ledger Technology	Theory Only	1.0	2	0	0	0	2.0
34	BCSE329P	Blockchain and Distributed Ledger Technology Lab	Lab Only	1.0	0	0	2	0	1.0
35	BCSE330L	Public Key Infrastructure and Trust Management	Theory Only	1.0	3	0	0	0	3.0
36	BCSE331L	Exploratory Data Analysis	Theory Only	1.0	2	0	0	0	2.0
37	BCSE331P	Exploratory Data Analysis Lab	Lab Only	1.0	0	0	2	0	1.0
38	BCSE332L	Deep Learning	Theory Only	1.0	3	0	0	0	3.0
39	BCSE332P	Deep Learning Lab	Lab Only	1.0	0	0	2	0	1.0
40	BCSE333L	Statistical Inference	Theory Only	1.0	2	0	0	0	2.0
41	BCSE333P	Statistical Inference Lab	Lab Only	1.0	0	0	2	0	1.0
42	BCSE334L	Predictive Analytics	Theory Only	1.0	3	0	0	0	3.0
43	BCSE335L	Healthcare Data Analytics	Theory Only	1.0	3	0	0	0	3.0
44	BCSE336L	Financial Data Analytics	Theory Only	1.0	2	0	0	0	2.0
45	BCSE336P	Financial Data Analytics Lab	Lab Only	1.0	0	0	2	0	1.0
46	BCSE391J	Technical Answers to Real Problems Project	Project	1.0	0	0	0	0	3.0

Discipline Elective									
47	BCSE392J	Design Project	Project	1.0	0	0	0	0	3.0
48	BCSE393J	Laboratory Project	Project	1.0	0	0	0	0	3.0
49	BCSE394J	Product Development Project	Project	1.0	0	0	0	0	3.0
50	BCSE396J	Reading Course	Project	1.0	0	0	0	0	3.0
51	BCSE397J	Special Project	Project	1.0	0	0	0	0	3.0
52	BCSE398J	Simulation Project	Project	1.0	0	0	0	0	3.0
53	BCSE401L	Internet of Things	Theory Only	1.0	3	0	0	0	3.0
54	BCSE402L	Big Data Analytics	Theory Only	1.0	3	0	0	0	3.0
55	BCSE403L	Digital Image Processing	Theory Only	1.0	3	0	0	0	3.0
56	BCSE404L	Internet and Web Programming	Theory Only	1.0	3	0	0	0	3.0
57	BCSE405L	Advanced Java Programming	Theory Only	1.0	3	0	0	0	3.0
58	BCSE406L	NoSQL Databases	Theory Only	1.0	3	0	0	0	3.0
59	BCSE407L	Computer Vision	Theory Only	1.0	3	0	0	0	3.0
60	BCSE408L	Cloud Computing	Theory Only	1.0	3	0	0	0	3.0
61	BCSE409L	Natural Language Processing	Theory Only	1.0	3	0	0	0	3.0
62	BCSE410L	Cyber Security	Theory Only	1.0	3	0	0	0	3.0
63	BCSE411L	Robotics and Automation	Theory Only	1.0	3	0	0	0	3.0
64	BCSE412L	Parallel Computing	Theory Only	1.0	3	0	0	0	3.0
65	BCSE413L	Soft Computing	Theory Only	1.0	3	0	0	0	3.0
66	BCSE414L	High Performance Computing	Theory Only	1.0	3	0	0	0	3.0
67	BCSE431L	Fundamentals of Quantum Computing	Theory Only	1.0	3	0	0	0	3.0
68	BCSE432E	Reinforcement Learning	Embedded Theory and Lab	1.0	3	0	2	0	4.0
69	BECM302E	Augmented and Virtual Reality	Embedded Theory and Lab	1.0	3	0	2	0	4.0
70	BEEE303L	Control Systems	Theory Only	1.0	3	0	0	0	3.0
71	BEEE303P	Control Systems Lab	Lab Only	1.0	0	0	2	0	1.0

Projects and Internship									
sl.no	Course Code	Course Title	Course Type	Ver sio n	L	T	P	J	Credits
1	BCSE399J	Summer Industrial Internship	Project	1.0	0	0	0	0	1.0
2	BCSE497J	Project - I	Project	1.0	0	0	0	0	3.0
3	BCSE498J	Project - II / Internship	Project	1.0	0	0	0	0	5.0
4	BCSE499J	One Semester Internship	Project	1.0	0	0	0	0	14.0

Open Elective									
sl.no	Course Code	Course Title	Course Type	Ver sio n	L	T	P	J	Credits
1	BCSE353E	Information Security Analysis and Audit	Embedded Theory and Lab	1.0	1	0	2	0	2.0

Open Elective									
2	BCSE354E	Information Security Management	Embedded Theory and Lab	1.0	1	0	2	0	2.0
3	BHUM111L	Happiness and Well-being	Theory Only	1.0	3	0	0	0	3.0
4	BHUM201L	Mass Communication	Theory Only	1.0	3	0	0	0	3.0
5	BHUM202L	Rural Development	Theory Only	1.0	3	0	0	0	3.0
6	BHUM203L	Introduction to Psychology	Theory Only	1.0	3	0	0	0	3.0
7	BHUM204L	Industrial Psychology	Theory Only	1.0	3	0	0	0	3.0
8	BHUM205L	Development Economics	Theory Only	1.0	3	0	0	0	3.0
9	BHUM206L	International Economics	Theory Only	1.0	3	0	0	0	3.0
10	BHUM207L	Engineering Economics	Theory Only	1.0	3	0	0	0	3.0
11	BHUM208L	Economics of Strategy	Theory Only	1.0	3	0	0	0	3.0
12	BHUM209L	Game Theory	Theory Only	1.0	3	0	0	0	3.0
13	BHUM210E	Econometrics	Embedded Theory and Lab	1.0	2	0	2	0	3.0
14	BHUM211L	Behavioral Economics	Theory Only	1.0	3	0	0	0	3.0
15	BHUM212L	Mathematics for Economic Analysis	Theory Only	1.0	3	0	0	0	3.0
16	BHUM213L	Corporate Social Responsibility	Theory Only	1.0	3	0	0	0	3.0
17	BHUM214L	Political Science	Theory Only	1.0	3	0	0	0	3.0
18	BHUM215L	International Relations	Theory Only	1.0	3	0	0	0	3.0
19	BHUM216L	Indian Culture and Heritage	Theory Only	1.0	3	0	0	0	3.0
20	BHUM217L	Contemporary India	Theory Only	1.0	3	0	0	0	3.0
21	BHUM218L	Financial Management	Theory Only	1.0	3	0	0	0	3.0
22	BHUM219L	Principles of Accounting	Theory Only	1.0	3	0	0	0	3.0
23	BHUM220L	Financial Markets and Institutions	Theory Only	1.0	3	0	0	0	3.0
24	BHUM221L	Economics of Money, Banking and Financial Markets	Theory Only	1.0	3	0	0	0	3.0
25	BHUM222L	Security Analysis and Portfolio Management	Theory Only	1.0	3	0	0	0	3.0
26	BHUM223L	Options , Futures and other Derivatives	Theory Only	1.0	3	0	0	0	3.0
27	BHUM224L	Fixed Income Securities	Theory Only	1.0	3	0	0	0	3.0
28	BHUM225L	Personal Finance	Theory Only	1.0	3	0	0	0	3.0
29	BHUM226L	Corporate Finance	Theory Only	1.0	3	0	0	0	3.0
30	BHUM227L	Financial Statement Analysis	Theory Only	1.0	3	0	0	0	3.0
31	BHUM228L	Cost and Management Accounting	Theory Only	1.0	3	0	0	0	3.0
32	BHUM229L	Mind, Embodiment and Technology	Theory Only	1.0	3	0	0	0	3.0
33	BHUM230L	Health Humanities in Biotechnological Era	Theory Only	1.0	3	0	0	0	3.0
34	BHUM231L	Reproductive Choices for a Sustainable Society	Theory Only	1.0	3	0	0	0	3.0
35	BHUM232L	Introduction to Sustainable Aging	Theory Only	1.0	3	0	0	0	3.0
36	BHUM233L	Environmental Psychology	Theory Only	1.0	3	0	0	0	3.0
37	BHUM234L	Indian Psychology	Theory Only	1.0	3	0	0	0	3.0
38	BHUM235E	Psychology of Wellness	Embedded Theory and Lab	1.0	2	0	2	0	3.0
39	BMGT108L	Entrepreneurship	Theory Only	1.0	3	0	0	0	3.0
40	BSTS301P	Advanced Competitive Coding - I	Soft Skill	1.0	0	0	3	0	1.5
41	BSTS302P	Advanced Competitive Coding - II	Soft Skill	1.0	0	0	3	0	1.5
42	CFOC133M	E-Business	Online Course	1.0	0	0	0	0	3.0

Open Elective									
43	CFOC191M	Forests and their Management	Online Course	1.0	0	0	0	0	3.0
44	CFOC448M	Applied Ergonomics	Online Course	1.0	0	0	0	0	3.0
45	CFOC599M	Leadership and Team Effectiveness	Online Course	1.0	0	0	0	0	3.0
46	CFOC642M	Conservation Economics	Online Course	1.0	0	0	0	0	3.0
47	CFOC647M	Air pollution and Control	Online Course	1.0	0	0	0	0	3.0
48	CFOC651M	Psychology of Stress, Health and Well-being	Online Course	1.0	0	0	0	0	3.0
49	CFOC736M	Organizational Behaviour: Individual Dynamics in Organization	Online Course	1.0	0	0	0	0	3.0

Bridge Course									
sl.no	Course Code	Course Title	Course Type	Ver sio n	L	T	P	J	Credits
1	BENG101N	Effective English Communication	Lab Only	1.0	0	0	4	0	2.0

Non-graded Core Requirement									
sl.no	Course Code	Course Title	Course Type	Ver sio n	L	T	P	J	Credits
1	BCHY102N	Environmental Sciences	Online Course	1.0	0	0	0	0	2.0
2	BCSE101N	Introduction to Engineering	Project	1.0	0	0	0	0	1.0
3	BHUM101N	Ethics and Values	Online Course	1.0	0	0	0	0	2.0
4	BSSC101N	Essence of Traditional Knowledge	Online Course	1.0	0	0	0	0	2.0
5	BSSC102N	Indian Constitution	Online Course	1.0	0	0	0	0	2.0